

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA43

Name: P. Harika

Register Number: 192311243

P. Harika

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: Total Marks: 10

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.

- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

- A. `<media src="video.mp4"></media>`
- B. `<video src="video.mp4" controls></video>`
- C. `<movie src="video.mp4"></movie>`
- D. `<vid src="video.mp4"></vid>`

Answer: B. `<video src="video.mp4" controls></video>`

Justification:

- The `<video>` element is the standard HTML5 tag used to embed videos.
- The `controls` attribute provides built-in play, pause, volume, and fullscreen controls.
- It is supported by all modern browsers.

Why the others are incorrect:

- **A.** `<media>` - Not a valid HTML element.
- **C.** `<movie>` - Not a valid HTML element.
- **D.** `<vid>` - Not a valid HTML element.

Snippet 2: Form Submission Methods

- A. GET
- B. POST
- C. PUT
- D. DELETE

Answer: B. POST

Explanation:

- **POST** is used to send user data securely to the server.
- Form data is sent in the request body instead of the URL.
- It is suitable for registration forms because it can transmit sensitive information like passwords.

Why the others are incorrect:

- **GET** - Sends data in the URL; not suitable for sensitive information.
- **PUT** - Primarily used to update existing resources.
- **DELETE** - Used to delete resources.

Snippet 3: Navigation Elements

- A. `<div>`
- B. `<nav>`

- C. <section>
- D. <header>

Answer: B. <nav>

Justification:

- The <nav> element is specifically designed to contain navigation links. • It improves website accessibility and semantic structure.
- Search engines and screen readers recognize it as the navigation section.

Why the others are incorrect:

- <div> - Generic container with no semantic meaning.
- <section> - Represents a thematic section, not specifically navigation.
- <header> - Represents introductory content and may contain navigation, but it is not exclusively for navigation.

Snippet 4: XML Syntax

- A. <note><to>User</to><from>Admin</from>
- B. <note><to>User</to><from>Admin</from></note>
- C. <note><to>User</from></note>
- D. <note to="User"></note>

Answer: B.

```
<note><to>User</to><from>Admin</from></note>
```

Explanation:

- It is the correct XML because all tags are properly opened and closed.

Errors in other options:

- **A:** Missing </note> closing tag.
- **C:** Mismatched closing tag (</from> instead of </to>).
- **D:** Missing > at the end of </note>.

Snippet 5: Web Communication Protocols

- A. FTP
- B. HTTP
- C. SMTP
- D. SNMP

Answer: B. HTTP

Justification:

- **HTTP (HyperText Transfer Protocol)** is the standard protocol used for communication between web browsers and web servers.

- For secure communication, websites typically use **HTTPS**, which is HTTP over SSL/TLS.

Why the others are incorrect:

- **FTP** - Used for file transfer.
- **SMTP** - Used for sending emails.
- **SNMP** - Used for network management and monitoring.

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

1. Correct HTML tag for video embedding

Answer: B. `<video src="video.mp4" controls></video>`

Justification: The `<video>` tag is the standard HTML5 element for embedding videos and provides playback controls.

2. Appropriate HTTP method for form submission

Answer: B. POST

Explanation: POST sends form data in the request body, making it suitable for registration forms and sensitive information.

3. Proper semantic element for navigation

Answer: B. `<nav>`

Justification: The `<nav>` element is specifically used to define navigation links, improving accessibility and page structure.

4. Correct XML syntax

Answer: B. `<note><to>User</to><from>Admin</from></note>`

Explanation: It is well-formed XML with all tags properly opened and closed.

Errors:

- **A:** Missing </note> tag.
- **C:** Incorrect closing tag (</from> instead of </to>).
- **D:** Missing > in the closing tag.

5. Correct web communication protocol

Answer: B. HTTP

Justification: HTTP is the standard protocol used for communication between web browsers and web servers.

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case 25	Thorough understanding ; clearly identifies all	Good understanding ng with	Basic understanding ng ; some	Poor understanding; problem not identified correctly

		most key issues issues identified	issues missed	
Application of Concepts 25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts

Analysis 20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design 20	Innovative, feasible	Logical solutions	Basic solutions with weak	Incorrect or no solution provided

	solutions with strong justification with adequate justification justification	
Clarity 10	Clear, well structured, and easy to understand Organized and understanda ble presentation Limited clarity and structure	Poor clarity; difficult to understand